Qian 2020 Breast Cancer — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 7 patients traverse one continuous manifold: the shared developmental axis (signature PC1, 54% var) is spanned by 7/7 patients (minor samples alone cover 78%); module variance from patient identity only R^2=0.05–0.15. iLISI is imbalance-limited by the 65%-dominant tumour (ceiling 2.26 of 7); the call is led by the embedding + biology, not a scalar.

Gate 2 — module geography → GO (non-branching)

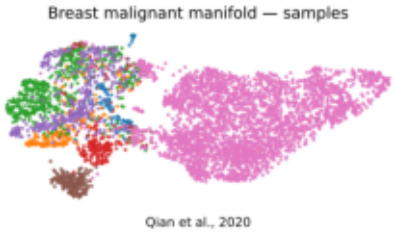
The embryonic programs form one developmental geography: a spatial Gast↔Crest inverse gradient (smoothed r = −0.49, batch-confirmed within patients), Gast & pSom overlapping toward the origin. An honest PPT tree put every tip at negative Gast — no Gast arm, no Gast↔pSom hub. Crest is an inverse gradient, not a Gast-arm enrichment: no inversion flag.

Gate 3 — hub cross-repression → NA

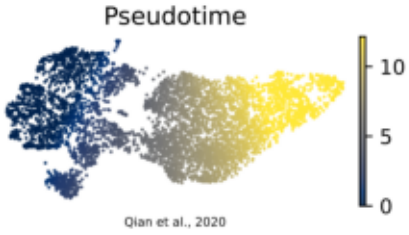
Non-branching manifold — no bifurcation and no Gast/pSom hub, so the cross-repression test does not apply.

STEP FIGURES

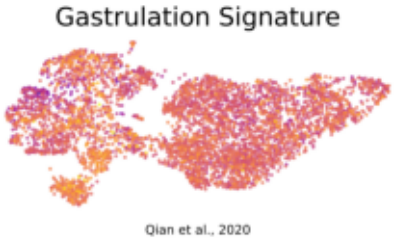
Manifold — samples



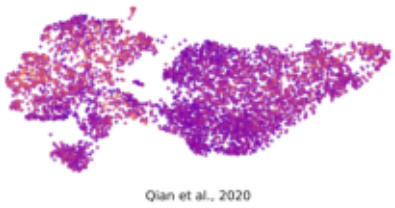
Pseudotime



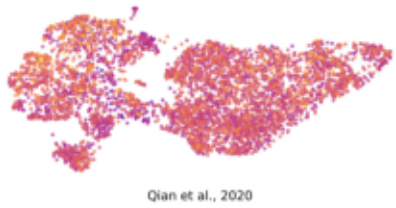
Gastrulation



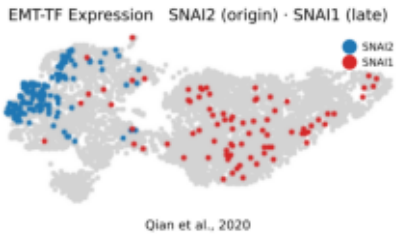
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Qian et al., 2020), 10x — 4,801 malignant cells, 7 patients (treatment-naive).

Verdict.
The embryonic programs reproduce as a NON-BRANCHING manifold: gastrulation and somitogenesis overlap toward a developmental origin, with neural crest forming a spatial inverse gradient to gastrulation (Crest↔Gast r = −0.49) and no Gast/pSom hub or bifurcation. Geometry = Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TF axis SNAI2 (early/origin) → SNAI1 (late), detection-gated panel.

Parameter choices

Malignant cells	4,818 raw → 4,801 (thresholded, de-islanded)	**Sample threshold**	≥100 cells (gap rule mis-fires: dominant high-side cliff) → 7/7
Overdispersed genes	1,225 (scFates find_overdispersed)	**Integration**	Harmony θ=10, n_pcs=20 (iLISI-sweep derived)
Embedding	UMAP spread 1.3, n_neighbors 30, seed 1; island removal 17 cells / 1 speck	**Trajectory**	ElPiGraph principal CURVE, 20 nodes (fold-free, self-approach 2.99)
Root	Gast/pSom-high, Crest-low tip (Crest late)	**EMT-TFs**	SNAI2 (origin) · SNAI1 (late), detection-gated